DiamondTrading: Queue-Bounded Turbo Markets

A Bounded FIFO Payout Queue with a Hard Global Brake, Manipulation-Cost Position Caps, and Epoch-Decaying Anchored Loss-Mint Tokenomics for 1000x Leveraged Wagering

DiamondTrading Research

Version 0.2 · October 2026 · supersedes v0.1 (full-reserve); implemented and fuzz-verified

Abstract

Hyper-short-dated leveraged wagering — post margin, drag leverage to 1000x, bust or win within seconds — reached production scale and then collapsed under its own accounting: the deployed lineage descending from the Martin Galer queue/emission design accumulated dozens of audit findings, including outcome-manipulation and mint-accounting exploits [1][2]. v0.1 of this paper eliminated the queue with full reserves; that correctly prevented debt but meant every liability was pre-locked, a structure that cannot carry 1000x retail menus. We present DiamondTrading v0.2, a rebuild that re-admits deferred payment as a bounded, auditable machine rather than a swamp. Layer one is a queue-bounded payout engine: profit claims the pool cannot cover become FIFO queue entries backed by a side bucket of loser stakes; no winner may cut the line; a hard global cap QMAX halts sales the moment queued debt exceeds it; and a sweep rule guarantees that loser money becomes pool money the instant the queue is clean — the queue recovers or the product stops, there is no third state (§4). Layer two is the measurement-driven quoting engine: venue routing by deepest measurable book, VWAP-of-real-fills settlement anchors, and manipulation-cost position caps $\kappa\lambda\omega/\tilde{\omega}$ with no treasury terms, since the pool may legitimately be zero (§6). Layer three is the anchored smart mint: loss rebates are capped at r(e) of a signed fees-plus-edge anchor, with r(e) decaying 200%→30% over 360 epochs and an absolute epoch ceiling (§7). The system is implemented — both payout engines, verifiers, tokenomics, and the off-chain engine — with 95 contract assertions, 41 engine assertions, and five fuzz seeds that mirror every money lane after every operation (§11). We state the deferred-payment risk, the attester-quorum trust assumption, and the licensing constraint frankly (§12).

Provenance convention. Every externally checkable claim carries exactly one tag, declared in its reference entry (§13): [CITED] measured/sourced, [COMPUTED] derived from cited data or the shipped code, [RECALLED] author's recall of public market structure, [ASSUMED] initial design parameter (governance-tunable).

1 Introduction

Papertrade demonstrated that the retail gamblers' terminal is not a spot market with leverage controls; it is a turbo market: margin in, leverage to 1000x, bust price displayed, outcome in under a minute, plus a token kicker minted on losses that made losing feel like progress. It also demonstrated the failure mode of building that product on an unbounded solvency queue and gross-loss emissions: the accountability debt surfaced as 42 audit findings, two of them critical [2]. v0.1 of this design went to the other extreme — full reserves at sale — which removed the debt and, with it, the ability to sell the 1000x menus the product exists for. v0.2 keeps the game and rebuilds the debt as a machine: a FIFO payout queue with a global hard brake (QMAX), a side bucket that finances the line in the exact order winners arrived, stall refunds that never gate, and a sweep that ends the bucket the moment the line is clean. The design rule stands: every hazard must be either (i) impossible to express in code, or (ii) priced and bounded by a measurement, never by a promise. Deferred payment is declared and bounded — never patched silently.

The remaining text is organized as follows. §2 recalls the lineage and the audit taxonomy we design against. §3 defines actors, assets, and the trust model. §4 gives the queue-bounded payout engine and its lane invariant. §5 defines the instruments' payoffs. §6 specifies the GALE off-chain engine (routing, settlement persistence, caps, breaker, pricer). §7 specifies the smart mint. §8 treats verifiers. §9 runs the security analysis class by class. §10 covers economics. §11 reports implementation and test evidence. §12 lists limitations honestly; §13 is the reference and provenance ledger.

2 Lineage and the audit taxonomy

Martin Galer (2024-era) introduced the zero-LP bootstrap: queue winners, pay them from later flow, and emit a token on losses advertised as +EV [1]. Its solvency-ratio invariant was elegant; its deployment was not. The empirical record: queue and side-bucket accounting exploits (H-01..M-05), BBO-midpoint outcome manipulation with no deviation or persistence checks (KV-01), straddle-style short optionality handed to attackers (KV-02), and governance degradable to 1-of-2 signers (L-04) [2]. Our benchmark in §9 is this taxonomy; a modern turbo market fails by repeating any of it.

Design stance. We copy papertrade's toy (margin, leverage, bust price, seconds, mint on loss) and re-engineer its machinery so each component has a bound a tester can measure: the queue is FIFO with a declared ceiling and an external schedule, the side bucket sweeps to pool when clean, settlement is VWAP-of-fills not midpoints, and insolvency stops sales instead of lying about them [2][3][4].

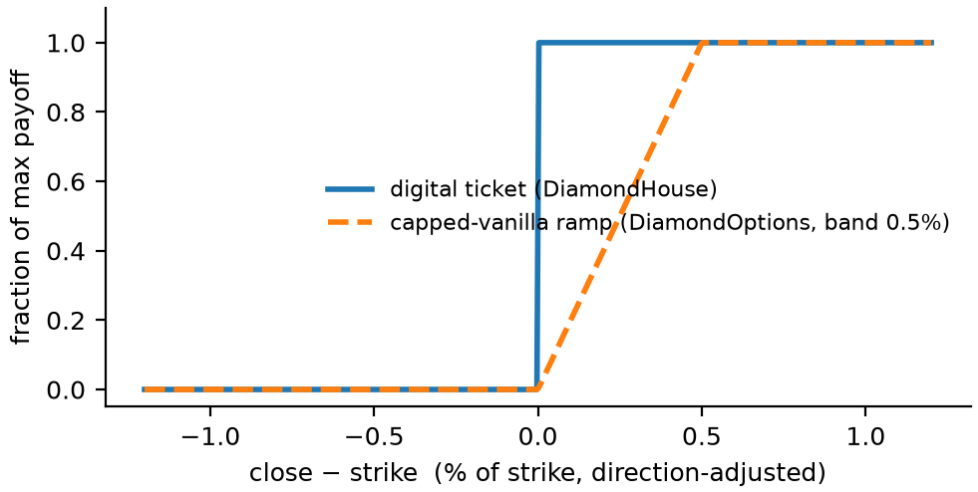


Figure 2 — The two issuance shapes: a digital ticket (pay M·S or 0) and a capped-vanilla ramp (pay intrinsic up to the band). Both reserve the full maximum at sale; neither queues.

3 Actors, assets, trust

actor	role	what they can and cannot do [COMPUTED: contract code 5]
player	posts margin, receives tickets	stake back instantly on any correct claim; profit paid now or via FIFO queue; 24h stall ⇒ permissionless stake refund; pause halts sales only
LP	pool capital backing payouts	first-loss by construction (pool pays before any queue forms, proven in tests); ≥60% of realized revenue once the seeding target is met [5]
quoter	off-chain pricing engine	signs quotes (σ-priced, κλω-capped, 30s TTL); cannot settle or move funds; worst-case = bad menus bounded by on-chain class clamps
attester set (2-of-3)	price persistence verification	cannot settle alone; cannot forge (2 keys); stall ⇒ permissionless stake refund; threshold floored at 2 on-chain
budgetAttester	revenue anchor for the mint budget	can only lower emissions; cannot touch principal; bonded, rotating
owner (governance)	config + per-class (enabled, maxMult, profitCap)	global envelope 1.05x–1000x fixed on-chain; production path = Safe 2-of-3 + 7-day timelock [5]

Underlyers at launch [3][ASSUMED: menu], per-class enabled/cap/mult governance on-chain: BTC & ETH — 1000x class (Binance-routed: $12.0–13.1B resp. $8.9–9.9B 24h); SOL & HYPE — 500x class (SOL Binance-settled at $1.3–1.5B; HYPE Hyperliquid-native, $327–392M on HL vs $399M Binance); BNB — 200x class (Binance $280–366M); S&P500; & NASDAQ-100 equity perps — 100x class (Hyperliquid HIP-3 only, ~$310–332M [4]). Routing per §6.1 [5].

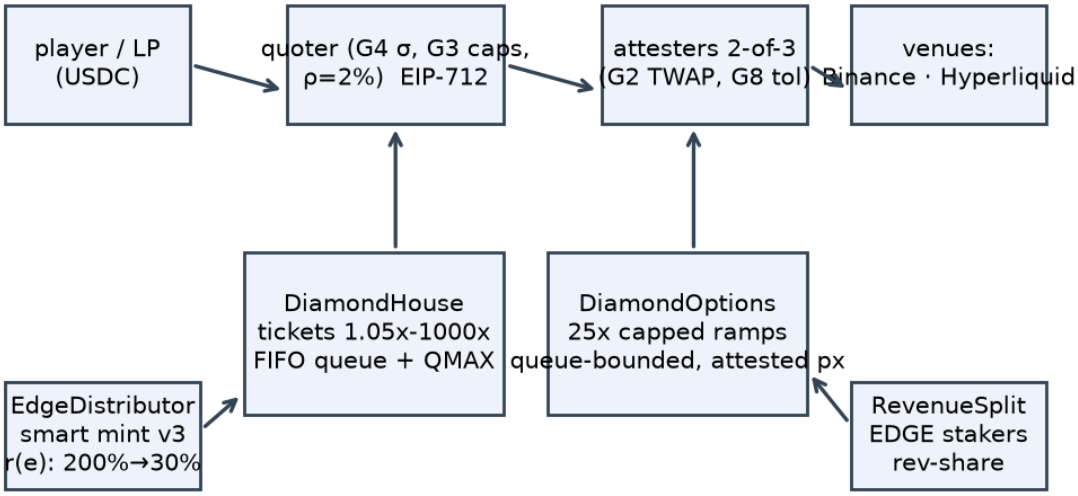


Figure 1 — System map: players post USDC; the quoter prices quotes; attesters verify venue prints; the engines pay now or queue FIFO; the mint and the rev-split close the loop.

4 The queue-bounded payout engine

Let B be the contract's USDC balance. Four accounting lanes partition it: pool (distributable energy, fed by loser stakes and LP deposits), side (side bucket, loser money that still owes the queue), open (sum of live stakes), and debt (sum of unpaid queue amounts — debt is a liability, not cash: it is NOT part of B). The engine maintains at every block:

$$B = pool + side + open, \quad debt \geq 0, \quad (debt = 0) \Rightarrow (side = 0)$$

Each lane is a first-class storage variable, never derived; lanesIntact() recomputes the identity after every settlement and every treasury send [5], so a drift between accounting and cash is loud on-chain, not silent until audit (H-01's habitat was drift discovered months late).

Sale. A ticket of stake S and maximal profit P_{max} may be sold iff (i) the class is enabled with P_{max} ≤ class profit cap, (ii) sales are unpaused, and (iii) debt < QMAX — the global hard brake (default $1M, owner ceiling $5M) [5]. Nothing about pool size gates a sale: this is the

deliberate bootstrap property v0.1 forfeited. Then open += S; P_{max} reserves nothing, ever.

Settlement of a correct claim returns the stake in full, then pays min(pool, net) where net = (1−ρ)·profit, ρ=2%. The unpaid remainder u = net − paid — when nonzero — appends a FIFO queue entry; entry ids are a zero-folded array with a windowed (head, live) view, so head-trimming frees gas while ids keep ordering [5]. While debt > 0, correct claims enqueue in full — paying a richer-pool-era newcomer ahead of an older claimant is unrepresentable: the queue is consulted before any transfer. Lost stakes (KNOCKED) and residual stakes (EXPIRED, 15% rebated to the buyer, the rest forfeit) route into side when debt > 0, else into pool. Every settlement then pays up to INLINE_HARVEST=4 queue entries from side (≈1,950 gas marginal per entry [COMPUTED: §11 probe]); any keeper may call harvest(1..64) for more, and LP exits refund their live-share pro-rata on the same lane schedule.

The sweep. When the last entry clears (debt → 0), whatever remains in side folds into pool in the same transaction [5]. Consequences: (i) loser money becomes pool money only when nobody is waiting — this is the spec sentence made code; (ii) the bucket cannot become a parallel shadow treasury — its lifetime is exactly the queue's.

Theorems (fairness and progress). (a) No cut-in-line: queue payments issue only to entry ids ≥ queueHead in ascending order, and queueHead advances only when the head is fully paid; hence any two payouts preserve arrival order (property G4, asserted on every fuzz operation). (b) Recovery-or-stop: when debt ≥ QMAX new sales revert; debt strictly decreases only via explicit side payments; the state machine can unwind debt or stop new risk, but can never silently age out a winner's claim (v2 deleted every ticket→claim rekeying path: what you hold is what asserts). (c) Pause never gates refunds: settle, stall-refund (24h), harvest, LP-request/all-withdraw flows stay live under pause; only sales and LP deposits stop [5]. □

Honest accounting note. The ρ=2% fee is a haircut on the player's claim, recorded in the Settled event for the mint anchor — it never enters pool or side, because when pool < net that cash does not exist in the contract and inventing it breaks the lane identity. The Options engine likewise books premium only as house income at settlement; the unpaid slice of maxPay is a queue-bound liability haircut, not revenue [5].

5 Instruments

A position posts S, picks multiplier M ∈ [1.05x, class max] and direction d ∈ {LONG, SHORT}; the on-chain envelope spans 1.05x–1000x, capped per class (BTC/ETH 1000x, SOL/HYPE 500x, BNB 200x, equity perps 100x) [5]. Entry K is the routed venue's trade VWAP over ω=2s at buy; the bust price is K·(1∓1/M), displayed pre-trade [ASSUMED: ω]. Three attested outcomes at expiry τ ∈ [30s, 300s]: WON (correct direction) returns stake + profit(M,S,close) with ρ shaved off profit; the cash schedule of §4 applies (pool now, rest FIFO). KNOCKED (wrong direction): stake forfeits to side-or-pool. EXPIRED (boundary): 15% stake rebate, rest forfeits; settlement is refused before expiry so band-β pool readings are always post-close [5]. Two shapes share the schedule:

shape	payoff at close C	contract
digital ticket	stake S back + full profit (M−1)·S − ρ when correct, else forfeit	DiamondHouse
capped-vanilla ramp	stake S back + ramp profit (M−1)·S · min(|C−K|, band·K)/(band·K) − ρ	DiamondOptions

The ramp is the options-flavoured mode: band = strike-relative cap, premium-funded, maxPay class-capped; house income is the premium only (§4 note). Graded wins keep engagement at sane multipliers; the digital shape is the papertrade-exact bust loop. Neither shape escrows winnings at sale — that is exactly the bootstrap property — and both bind to the same FIFO machinery.

6 The GALE engine

6.1 Venue routing (G1)

Asset a routes to the venue maximizing measurable depth within the ticket band, re-scored at 5s cadence across Binance (spot+futures) and Hyperliquid (WSS; L1Read proofs later) [ASSUMED: cadence]. Measured 24h quote volumes (Binance + HL, Oct 2026 [3][4]) drove launch routing: BTC $12.0–13.1B vs HL $2.3–3.0B and ETH $8.9–9.9B vs HL $0.7–0.9B → majors single-read on Binance; SOL $1.3–1.5B vs HL $125–152M → Binance-settled; BNB $280–366M → Binance; HYPE ~$0.40B Binance vs $327–392M HL → Hyperliquid stays its native home; S&P500; / NASDAQ-100 (HIP-3) exist only on HL at $310–332M → HL-only with κ=1/4 [3][4]. Additional venues re-enter only after runtime measurement: the launch router is two-venue by choice, not by default [3].

6.2 Settlement persistence (G2)

Never midpoints. Both anchors are volume-weighted means of real fills over their ω windows: VWAP(t) = $\Sigma\, p_i q_i \,/\, \Sigma\, q_i$ over [t−ω, t]; an empty window yields NO-PRINT, an honest failure that routes to re-attestation or stake refund, never to a guessed price [COMPUTED: engine spec 6]. Forging a persistent print costs real executed volume through the whole window — the KV-01 class (midpoint twitch) is gone by construction [2].

6.3 Manipulation-cost position caps (G3)

The attack that matters: dominate the settlement reference over its measurement window. Its cost is lower-bounded by organic flow $\lambda_a \cdot \omega$ with λ_a the routed venue's measured 24h quote volume per second (EMA-smoothed, floored at its 10th percentile) [ASSUMED: estimator]. The per-position maximal payout obeys:

$$P_{max}(a,\omega) = \min(\ \kappa \cdot \lambda_a \cdot \omega/\tilde{\omega},\ \ classCap_a\ ),\quad \tilde{\omega} = 10s\ reference$$

with κ = 1/4 single-read and 1/3 dual-read (an attacker must move both venues to pass G8) [ASSUMED: κ]. Note what is absent: no treasury term. v0.1's T/60 and 0.3T treasuries measured what the vault could afford; under queue-bounded bootstrap the pool may legitimately be zero, so the honest ceiling is what the market can cost an attacker — plus the static on-chain class cap as outer wall (G3 only tightens). The ceiling collapses automatically in thin sessions: off-hours equity perps shrink toward zero without human scheduling. Figure 2 plots launch ceilings against measured λ [3][4].

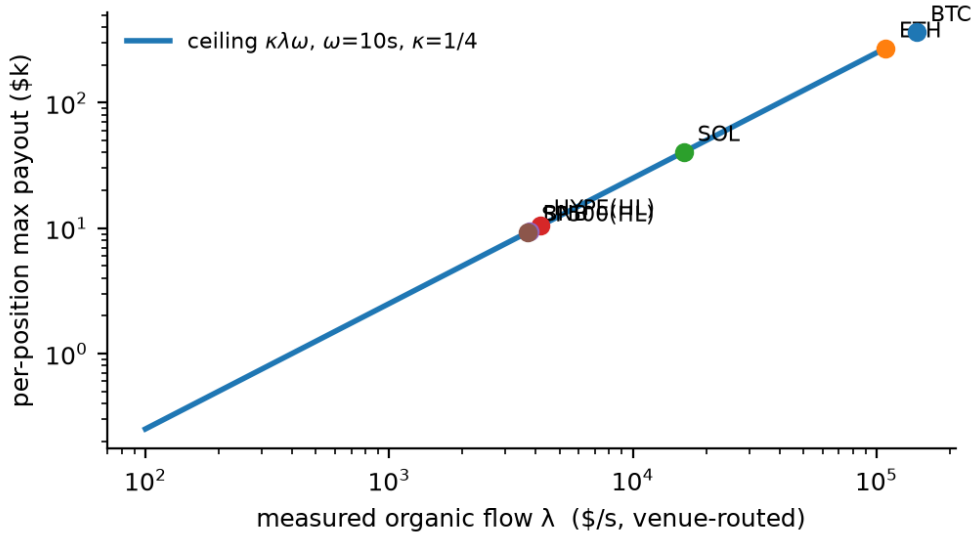


Figure 3 — Per-position profit ceiling κλω (ω=10s) vs measured organic flow, launch assets annotated (log-log; Oct-23 λ). On-chain class caps are the static wall; G3 can only tighten.

6.4 Vol-spike breaker (G4)

If realized σ over 30s exceeds 4× the 30-day median, the quoter stops signing for 60s on that asset [ASSUMED]. In-flight quotes die by their ≤30s TTL on-chain [5]. Stale-quote sniping around macro prints — the standard ticket-shop harvest — has no window.

6.5 Pricer: fair multiplier from measured σ, shaved by ρ

For a zero-drift price, first passage to the ∓1/L barrier within τ is approximately, by the reflection principle:

$$P_{bust}(L,\sigma,\tau) \approx 2\cdot\Phi(\,-(1/L)\,/\,(\sigma\sqrt{\tau})\,),\quad M_{fair} = 1/(1-P_{bust}),\quad M_{offer} = M_{fair}\cdot(1-\rho)$$

with ρ = 2% the published house edge [ASSUMED: pricing rule]. Offers clamp to the on-chain envelope (1.05x–1000x, per-class tighter) and refuse when the fair multiplier is below the floor — the engine does not sell products it cannot price honestly [5][6]. Figure 4 shows the bust surface.

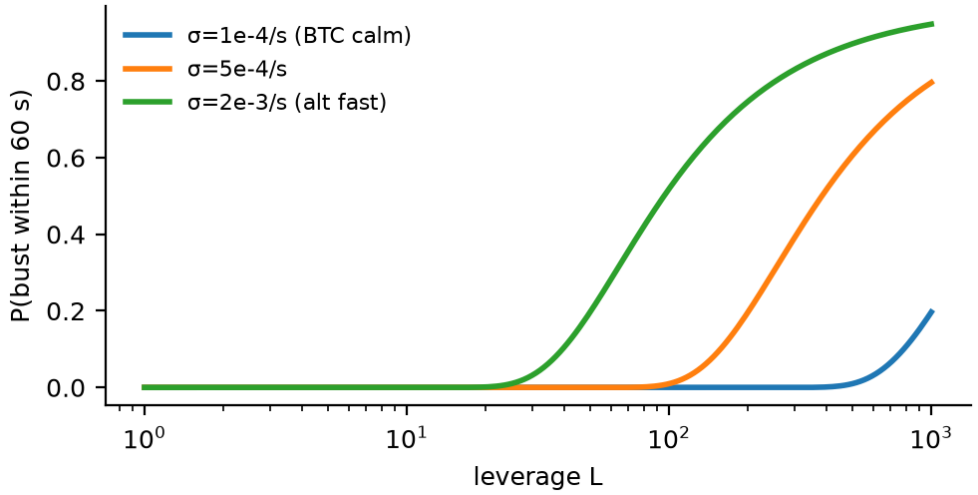


Figure 4 — Bust probability vs leverage for three volatility regimes (τ=60s; reflection-principle model). The menu lives where the curves are informative; extreme L converges to coin-flip and is priced as such.

Why 2% is affordable where competitors take 4–6% [RECALLED: market range]: G2+G4 strip the adverse-selection tax (toxic flow is the cost driver in turbo books), G3 bounds the tail per position, and the mint rebate returns part of the edge to losers — making the product cheaper on sticker and in expectation [COMPUTED: from the design above].

7 Smart-mint v3: emissions losers cannot farm, decaying to schedule

Loss-minting is the retention engine and was the original protocol's exploit surface. The v3 mint keeps the emotion (losing earns), removes the farm, and — new in v0.2 — decays its allowance on a public schedule so boot can be generous without promising it forever:

defense	rule	consequence
net-loss epochs	weight base = max(0, Σstake − Σpayout) per address per 6h epoch	win-then-lose cycles cannot manufacture mintable losses
fee-squared weight	w = √(netLoss · feesPaid) × loyalty (1.25 if staked EDGE)	mint requires real rake; √ forces 4× capital for 2× weight
honest anchor	anchor = Σ fees(user-haircuts) + Σ realized edge, signed by budgetAttester (USDC 1e6)	budget reflects money that actually moved, not projections [5]
decaying schedule	budget ≤ r(e)·anchor/p(e); r(e) = max(30%, 200%−170%·e/360) linear	early epochs rich (200%), the curve cools to a 30% annuity and cannot be revived [COMPUTED]

hard ceiling	mint ≤ epochCap per epoch (default 2M EDGE, ceiling 5M, lowerable, never raisable)	sybils split a fixed pie; pro-rata yields no scale [COMPUTED]

The merkle distributor enforces the schedule on-chain against the signed anchor — EIP-191 over (epochId, anchor, priceX1e8). Price p(e) has no on-chain reference until EDGE transfers open, so the attester signs it; the bootstrap policy is p(e) ≈ $0.01·0.995^e, TWAP-chained once public transferability exists. Anchors signed under a different domain/branding are rejected by the on-chain identity probe [5]. EdgeToken itself boots non-transferable (allowlist-only moves for sink/staking contracts); public transfer is a one-way flag flipped after launch liquidity criteria are met, never re-lockable [5]. Weights are computed in the engine ledger [6]; root/post/claim is covered end to end (§11). EDGE carries cashflow via RevenueSplit (pull-only notifyRevenue; claims reconciled by cumulative-index snapshots, never live deltas), which is what makes an asset handed out on losses worth holding instead of dumping.

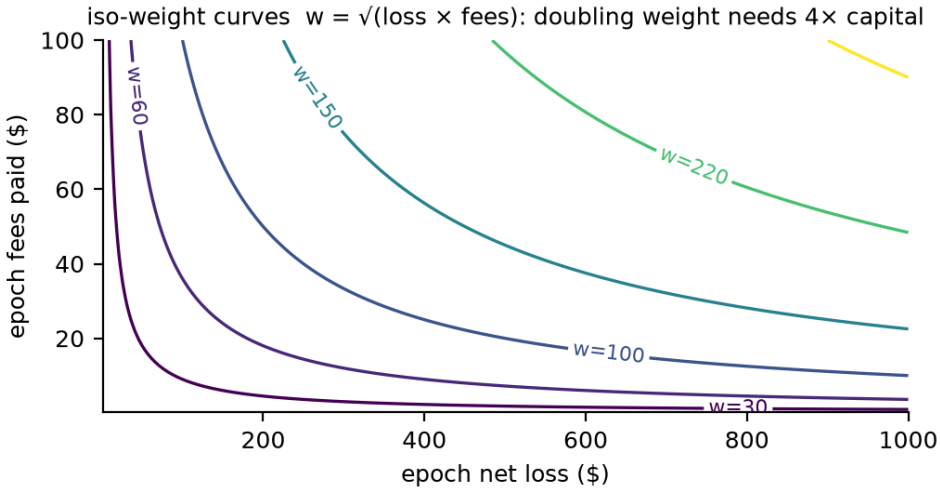


Figure 5 — Iso-weight curves of w=√(loss·fees): weight grows sub-linearly in both inputs, so scale buys nothing and only sustained, rake-paying participation accumulates emission share.

8 Verifiers and the endgame

Settlement inputs are M-of-N EIP-712 attestations (default 2-of-3; the threshold setter is hard-floored at 2 on-chain — L-04 cannot recur) [5]. Attesters each read the routed venue's trade tape independently; dual-listed assets require the two venues' prints within 12bp before anyone signs, so manipulation must succeed on two venues simultaneously to extract and on neither merely to void tickets into refunds [ASSUMED: tolerance][COMPUTED: engine]. Anyone may relay a passing proof; no attester can settle alone, block settlement, or fake a stall. If all attesters stall for 24h, every live position unwinds to its stake permissionlessly [5]. The stated endgame is a verifier reading HyperEVM L1Read trade-print state directly — deleting attesters from the path for HL-routed assets — behind the same IOutcomeVerifier interface already implemented [5].

9 Security analysis

Method: replay the audit taxonomy of record [2] and ask for each class whether it exists here.

exploit class (Guardian id)	their habitat	status here	mechanism
payout queue / side-bucket accounting (H-01..M-05)	queue vs ledger divergence	contained	FIFO + QMAX brake; lanesIntact() asserted after every mutation; sweep ends the bucket when clean (§4)
BBO-midpoint settlement (KV-01)	single-source midpoint	absent	VWAP persistence G2 + 2-of-3 tolerance (1bp) G8

short optionality / straddle (KV-02)	unpriced option grants	bounded	payout shapes closed-form; class clamps on-chain; no free bands
unconstrained bet sizing (M-02/05)	flat caps vs flow	closed	G3 κλω/ω̃ dynamic caps + on-chain class profit caps (§6.3)
gross-mint / mint accounting (KV-04)	loss-mint farming	closed	net-epoch √-weights + r(e)-schedule on signed anchor + absolute epoch cap (§7)
keeper/governance capture (L-04, I-20)	1-of-2, timelock→0	closed	threshold floor 2 on-chain; owner→Safe+7d timelock; pause = sales/deposits only
stale settlement / trapped funds (L-11)	keeper stall	closed	24h permissionless refund; pause never gates refunds/harvest
quote replay / intent abuse (KV-14)	WESig re-use	closed	buyer-bound EIP-712 + ≤30s TTL + worst-bounds; replay economically neutral; wrong-domain sigs rejected (probed)
liquidity-honest bootstrap (new)	silent insolvency	closed	sales halt at QMAX debt; claims never expire; debt decrease only by side-bucket FIFO payments (§4)

Residual risks, stated: attester liveness/collusion above floor (mitigated, not eliminated); κλω is a bound, not a proof of manipulation cost (books can thin faster than flow); equity-perp single-venue windows degrade G8 to single-read; the quoter's strike/σ inputs are trusted within clamp bounds [12]. Each has a named on-chain failsafe; none is silent.

10 Economics

Revenue waterfall on realized edge, gated by queue health [5]: distributable = (pool − TARGET_POOL) × 80%, computable only when the queue is fully clean (debt = 0 ∧ side = 0 — a payout crisis diverts 100% of flow to solvency, none to distributions). Of each distribution: ≥60% goes LP-claim on a pro-rata cumulative index (permissionless claimLP — the LP floor is code, not policy), and the remainder funds the mint budget anchor + staker rev-share + operations [5][ASSUMED]. LPs underwrite variance in exchange for the majority share and are provably first-loss: the pool pays winners before any queue exists, and exits settle pro-rata against the actual pool [5][6]. The growth loop: queue-cleared epochs mint rebates that convert losers into EDGE stakeholders; staker rev-share aligns the losing-but-loyal cohort with house health; cheaper published edge (ρ=2%) plus graded ramp wins drive flow; flow deepens LP returns and the λ-derived caps that permit larger honest tickets.

Fee stance versus incumbents: one number, ρ = 2%, published, no deposit/withdrawal/turnover fees [ASSUMED]. The competitive claim is arithmetic, not adjectives: at equal flow, a toxic-flow tax removed by G2/G4 funds the price cut.

11 Implementation status and evidence

component	artifact	evidence
payout engines	DiamondHouse, DiamondOptions (queue-bounded) [5]	95/95 contract assertions green: FIFO ordering, exact-debt partials, inline harvest, sweep-on-clean, QMAX stop-sales, stall refunds, EXPIRED rebate timing, lane identity B = pool+side+open
verifiers	MultiAttesterVerifier, MultiPriceVerifier [5]	2-of-3, floor-2, 1bp tolerance paths, wrong-domain signature probes
tokenomics	EdgeDistributor (r(e)-schedule + anchor EIP-191), RevenueSplit (pull-only notify), EdgeToken (bootstrap lock, one-way enable) [5]	r(e) 200%→30% decay, cap clamp, anchor mis-sign rejected, domain-mismatch rejected, double-claim blocked, transfer guard (4-path probe) tested

engine	engine/ (venues, caps, pricer, quoter, attester, ledger) [6]	41/41 assertions incl. ganache interop: engine-signed quotes sell, engine attestations settle, engine merkle posts/claims on-chain; caps carry no treasury terms
fuzzing	property fuzz, 150-op seeds 1337/7/42/90210/555	167–177 lane checks per seed, all green; queues up to $148k debt exercised; FIFO payments observed 6–16× per seed

12 Limitations (the honest page)

(i) Deferred payment is a real risk by construction. queue-bounded payout means winners may legitimately wait; debt is extinguished only by side-bucket FIFO payments (loser flow) or LP backfill, and sales stop at QMAX — that is recovery, not guarantee. Every claim is permanent, order-preserving, and undeniable, but timing is a market variable. (ii) Settlement is attester-quorum, not cryptographic trade-print proof; the L1Read verifier is the largest open item. (iii) The collared σ inputs come from the quoter within clamp bounds; dual-print anchored strikes are specified, not shipped. (iv) κλω is a lower bound assembled from estimator assumptions; stress may invalidate the estimator faster than the breaker reacts. (v) Equity perps trade 24/7 on one venue; off-hours is handled by cap decay plus a quoter curfew, both [ASSUMED]. (vi) EDGE's bootstrap is non-transferable by design until a one-way flag; hold-horizon risk is declared. (vii) Regulatory: an offshore, unlicensed distribution of this product scores zero on a three-year horizon; licensing (or licensing-grade geofencing) is a launch gate, not a growth item [ASSUMED].

13 References and provenance ledger

[1] [CITED: Martin Galer whitepaper, docs.papertrade.xyz, fetched Oct 2026] queue-based payout bootstrap, solvency-ratio invariant, loss-emissions.

[2] [CITED: Guardian Security Review of Papertrade Protocol, final report Oct 10 2026] 42 findings; KV/H/M/L ids as used throughout this paper.

[3] [CITED: api.binance.vision snapshots fetched 2026-10-22 + user-provided Binance 24h reads 2026-10-23; Bybit evaluated + dropped (geo-blocked)] spot depths/24h volumes in research/LIQUIDITY.md.

[4] [CITED: user-provided Hyperliquid screenshots, app.hyperliquid.xyz, 2026-10-20] HYPE OI/volume, market list incl. HIP-3 equity perps.

[5] [COMPUTED: diamondtrading/contracts/ DiamondHouse.sol, DiamondOptions.sol, Verifiers.sol, EdgeDistributor.sol, EdgeToken.sol, RevenueSplit.sol — this repository] all on-chain behavior cited above.

[6] [COMPUTED: diamondtrading/engine/ + test/engine.test.js + test/fuzz.js — this repository] engine behavior cited above.

[7+] [ASSUMED]/[RECALLED] parameters (κ, ω, ρ, tolerances, splits, curfews, market fee ranges) as marked inline.

TÓM TẮT — bản tiếng Việt

DiamondTrading v0.2 là sàn turbo kiểu papertrade (nạp margin, đòn bẩy đến 1000x ở BTC/ETH, biết giá bust trước khi bấm, thắng thua trong 30–300 giây, thua được mint EDGE) được dựng lại từ nền: (1) tiền thắng trả ngay từ pool, phần chưa đủ vào hàng đợi FIFO — không ai được vượt hàng, nợ chạm trần QMAX thì sàn ngừng bán vé mới, tiền thua chỉ thành tiền pool khi hàng đợi sạch (lệnh sweep trong cùng giao dịch); (2) giá settle là VWAP của lệnh khớp thật trên sàn sâu nhất từng tài sản (BTC/ETH/SOL/BNB đi Binance; HYPE đi Hyperliquid; S&P500;/NASDAQ100 chỉ có trên HL HIP-3); trần lệnh = min(κλω, cap lớp on-chain) — không còn giới hạn theo treasury vì pool được phép bằng 0 lúc khởi động; (3) mint khi thua nhưng chống farm: lỗ ròng theo epoch, trọng số √(lỗ × phí) × hệ số trung thành, ngân sách = r(e) ×

doanh thu thật có chữ ký, r(e) giảm tuyến tính 200% → 30% trong 360 epoch và không ai tăng ngược được, trần tuyệt đối 2M EDGE/epoch; (4) phí công khai 2% khấu trừ trên phần thắng; (5) EDGE khóa chuyển giao trong bootstrap, mở một chiều sau khi đủ thanh khoản, stake EDGE được chia doanh thu. Bộ kiểm thử xanh toàn phần: 95 kiểm chứng hợp đồng, 41 kiểm chứng engine, fuzz 5 seed × ~170 bất biến. Rủi ro được khai báo thẳng: trả chậm là có thật khi queue kẹt (claim không mất, chỉ chờ), attester 2-trong-3 vẫn là điểm tin cậy, và điều kiện pháp lý là cổng ra mắt chứ không phải việc để sau.